

Sustained Mask Mandates and High Mask-Wearing Behaviour During COVID-19: US State-Level Variation and Comparison with Australia

Part B final report

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1 Declaration

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2 Abstract

This study investigates the relationship between sustained mask mandate periods and high mask-wearing behaviour during the COVID-19 pandemic, focusing on differences among United states (US) states and further comparing the association between Australia and the US. This is significant because mask policy is not only a public health intervention but also reflects the link between policy environment, regional differences, and personal protective behaviour.

This study uses data from the YouGov COVID-19 Behaviour Tracker and the Oxford COVID-19 Government Response Tracker. Methods employed include US state-level descriptive comparisons, a US logistic regression model, and an Australia-US interaction regression model.

Results show that in most US states with sustained mandate start dates, the high mask-wearing rate during the sustained mandate period was higher than before the mandate period, but the magnitude of the change differed significantly between states. US logistic regression results show an association between sustained mask mandate periods and higher mask-wearing odds; the Australia-US interaction model further indicates that this association differs between the US and Australia.

Because this study uses observational survey data, the results should be interpreted as statistical associations rather than strict causal effects.

3 Introduction

During the COVID-19 pandemic, government mask policies were an important public health measure adopted by many countries to reduce the spread of the virus. However, whether a mask policy can increase mask-wearing rates depends not only on the policy itself, but also on factors such as local context, the duration of policy implementation, the strictness of the policy, and public attitudes. Previous studies have also shown that masks are not only a simple protective tool, but also have social, cultural, and political significance (Martinelli et al. 2020, 1).

Ryan et al. studied mask-wearing behaviour and protective health behaviours among Australians during the COVID-19 pandemic (Ryan et al. 2025). Their study examined which factors predict whether people will wear masks and whether these predictive factors changed before and after the implementation of mask mandates, which means government requirements to wear face coverings in specified public settings. Their results indicate that mask-wearing behaviour was associated with a variety of factors, including policy period, survey timing, state differences, perceived disease risk, trust in government. This suggests that mask-wearing behaviour is not only related to individual attitudes, but also to regional context and the state-level policy environment.

While Ryan et al. provided an important analysis of the situation in Australia, the association between mask mandates and mask-wearing behaviour may not be the same across all countries and regions. Previous research has shown that mask-wearing behaviour varies across cultural contexts (Lu, Jin, and English 2021, 1) and also varies across different spatial and temporal contexts in the US (Taube, Susswein, and Bansal 2023, 6–7). This means that findings from a study conducted in one country may not directly describe the relationship between policy and behaviour in other countries.

The motivation for choosing Australia and the United States (US) for comparison is that both are federal countries composed of different states. During the pandemic, state governments in both countries implemented mask policies at different times and with different levels of strictness. This state-level variation provides a useful research context for comparing the association between sustained mask mandate periods and high mask-wearing behaviour.

Based on this, this study addresses the following research questions (RQs):

- **RQ1:** How do high mask-wearing rates vary across US states before and during sustained mandate periods?
- **RQ2:** Is a sustained mask mandate period associated with high mask-wearing behaviour in the United States?
- **RQ3:** Does the association between sustained mandate periods and high mask-wearing behaviour differ between Australia and the United States?

If the comparison shows that Australia and the US have similar patterns, it suggests that the pattern found by Ryan et al. in Australia may be replicable and applicable across borders. If different patterns are found, it indicates that the relationship between mask policies and mask-wearing behaviour may depend on national and regional contexts. Because this study uses observational survey data and policy data, rather than randomized experimental data, the results can only be interpreted as association, not as causal effect.

4 Background

4.1 Literature review

During the COVID-19 pandemic, mask-wearing became not only a personal protective measure but also an important public health policy in many countries and regions (Martinelli et al. 2020, 1–2). Previous studies have shown that mandatory mask-wearing policies were associated with higher public mask-wearing rates. Näher et al. conducted a retrospective event study using data from 51 countries and found that mandatory mask-wearing increased, on average, self-reported mask-wearing rates (Näher et al. 2024, 1–2; 2024, 6). This suggests an important association between mask policies and individual mask-wearing behaviour.

Existing studies have shown that mask-wearing behaviour is associated with many factors, including social norms, collectivism, cultural context, self-efficacy, and risk perception. Lu et al. found that collectivist culture predicted mask use during COVID-19 (Lu, Jin, and English 2021, 1), while Schumpe et al. showed that social and community-level norms were the most generalisable predictor across several public health outcomes (Schumpe et al. 2022, 6). Yang et al. further argued that cross-country differences in mask-wearing cannot be explained only by epidemic severity, but also require attention to social norms, policy signals, and cultural factors (Yang et al. 2022, 1–2). Other studies also show that subjective norms, self-efficacy, perceived severity, perceived benefits, and perceived barriers can influence whether people adopt protective behaviours (Zhao and Knobel 2021, 1202; Jørgensen, Bor, and Petersen 2021, 679; White et al. 2022, 457).

In addition, mask-wearing behaviour also shows significant temporal and regional differences. Taylor et al. found that “pandemic fatigue” was characterised by a progressive decline over time in adherence to social-distancing guidelines and affected a minority of respondents in their US and Canadian sample (Taylor, Rachor, and Asmundson 2022, 1). MacIntyre et al. compared cities in Australia, the UK and the US and found that cities with mandatory mask-wearing policies generally had higher mask-wearing rates (MacIntyre et al. 2021, 199–200). Taube et al. used more than 8 million US survey responses and found that mask-wearing behaviour in the US was highly uneven in space, with urban areas generally having higher rates than rural areas, and mask-wearing rates changing significantly over time (Taube, Susswein, and Bansal 2023, 6–7). These studies suggest that analyses of mask-wearing policies should account for temporal and regional variation, rather than treating policy-behaviour associations as uniform across places and periods.

Overall, existing research has shown that mask-wearing behaviour is related to public health policy, but is also influenced by social, cultural, psychological, temporal and regional factors (Näher et al. 2024, 1–2; Lu, Jin, and English 2021, 1; Yang et al. 2022, 1–2; Jørgensen, Bor, and Petersen 2021, 679; Taylor, Rachor, and Asmundson 2022, 1; Taube, Susswein, and Bansal 2023, 6–7). Therefore, the association between mask mandates and mask-wearing behaviour may vary across countries and regions.

4.2 Statistical background

4.2.1 Logistic regression

Logistic regression is a statistical model commonly used for binary classification problems. Unlike linear regression, logistic regression does not directly model the probability. Instead, it first

transforms the probability to the logit scale and then establishes the relationship between the explanatory variables and the outcome variables. This ensures that the probabilities predicted by the model always lie between 0 and 1 (Sperandei 2014, 13–15).

The general form can be written as:

$$\text{logit} [\text{Pr}(Y_i = 1)] = \beta_0 + \boldsymbol{\beta}^\text{T} \mathbf{X}_i, \quad (1)$$

where Y_i is a binary outcome variable, with $Y_i = 1$ means that observation i belongs to the target category. $\mathbf{X}_i$ denotes a set of explanatory variables; β_0 is the intercept; and $\boldsymbol{\beta}$ is the vector of coefficients corresponding to the explanatory variables. A positive coefficient usually indicates that the variable is associated with increasing odds for the target category, whereas a negative coefficient indicates lower odds. To make the coefficients easier to interpret, logistic regression coefficients are often converted into odds ratios.

4.2.2 Interaction model

Interaction model are used to examine whether the association between one variable and an outcome changes depending on another variable.

In logistic regression, a general model with an interaction term can be written as:

$$\text{logit} [\text{Pr}(Y_i = 1)] = \beta_0 + \beta_1 A_i + \beta_2 B_i + \beta_3 (A_i B_i) + \boldsymbol{\beta}_X^\text{T} \mathbf{X}_i, \quad (2)$$

where A_i and B_i are two explanatory variables, $A_i B_i$ is their interaction term, and $\mathbf{X}_i$ represents other control variables. The interaction coefficient β_3 is used to assess whether the association between A_i and the outcome changes depending on B_i . If the interaction coefficient is different from 0, this suggests that the association may differ between different groups.

4.2.3 Key statistical terms

The logistic regression and interaction model results are interpreted using odds, odds ratios, 95% confidence intervals, p -values and interaction terms. These terms are used to describe the direction, strength and statistical uncertainty of the estimated associations. A summary of these terms is provided in Appendix A.

5 Methods

5.1 Australian Replication

During Australian replication, the analytical framework proposed by Ryan et al. (Ryan et al. 2025) was reproduced using the Australian YouGov COVID-19 Behaviour Tracker survey data (Jones 2020) together with policy data from the Oxford COVID-19 Government Response Tracker (OxCGRT) (Hale et al. 2021). The replication included data preprocessing, variables construction, and machine learning models to predict individual mask-wearing behaviour before and after the

implementation of the mask mandate, with the aim of comparing whether the key predictors changed between the two policy periods. The replication results were broadly consistent with those reported by Ryan et al.

5.2 Research Gap

Based on the Australian replication and literature reviewed above, existing studies have focused on either single-country analyses, broad cross-country comparisons, or selected city-level comparisons. Less attention has been given to whether sustained state-level mask mandate periods are associated with high mask-wearing behaviour in the US, and whether this association differs from the association observed in Australia.

5.3 Study Design

Based on this gap, this study extends the analysis to US data and further compares whether there are differences in the relationship between sustained mask mandate period and high mask-wearing behaviour between Australia and the US. This study focuses not on predictive model performance, but rather on using descriptive analyses and interpretable statistical models to investigate the adjusted association between policy periods and individual mask-wearing behaviour. Figure 1 summarises the overall analytical workflow.

5.4 Data Sources

5.4.1 YouGov COVID-19 Behaviour Tracker

This dataset contains individual-level COVID-19 behaviour survey information from 29 countries from March 2020 to March 2022, including demographic variables, health-related variables, protective behaviour items, mental health items, government trust-related variables, and survey dates. It comes from the YouGov COVID-19 Behaviour Tracker dataset, jointly released by Imperial College London and YouGov (Jones 2020). The Australian data comes from the cleaned and preprocessed Australian replication dataset in Part A. The US data comes from `united-states.csv`, which provides country-specific survey records for each participating country.

5.4.2 Oxford COVID-19 Government Response Tracker (OxCGRT)

The policy document used in this study is `OxCGRT_compact_subnational_v1.csv`, which records the COVID-19 mask policies and measures implemented in multiple countries and regions from January 2020 to December 2022, including lockdown policies, health policies, vaccine policies, and mask policies. It comes from the Oxford COVID-19 Government Response Tracker (OxCGRT) policy data (Hale et al. 2021).

Since the purpose of this study is to examine the association between mandatory mask orders and mask-wearing behaviour, this study focuses on policies related to face coverings (column identifier H6). H6M Facial Coverings represents government policy requirements for the use of masks or face coverings outside the home, and is an ordinal variable with values from 0 to 4, representing no

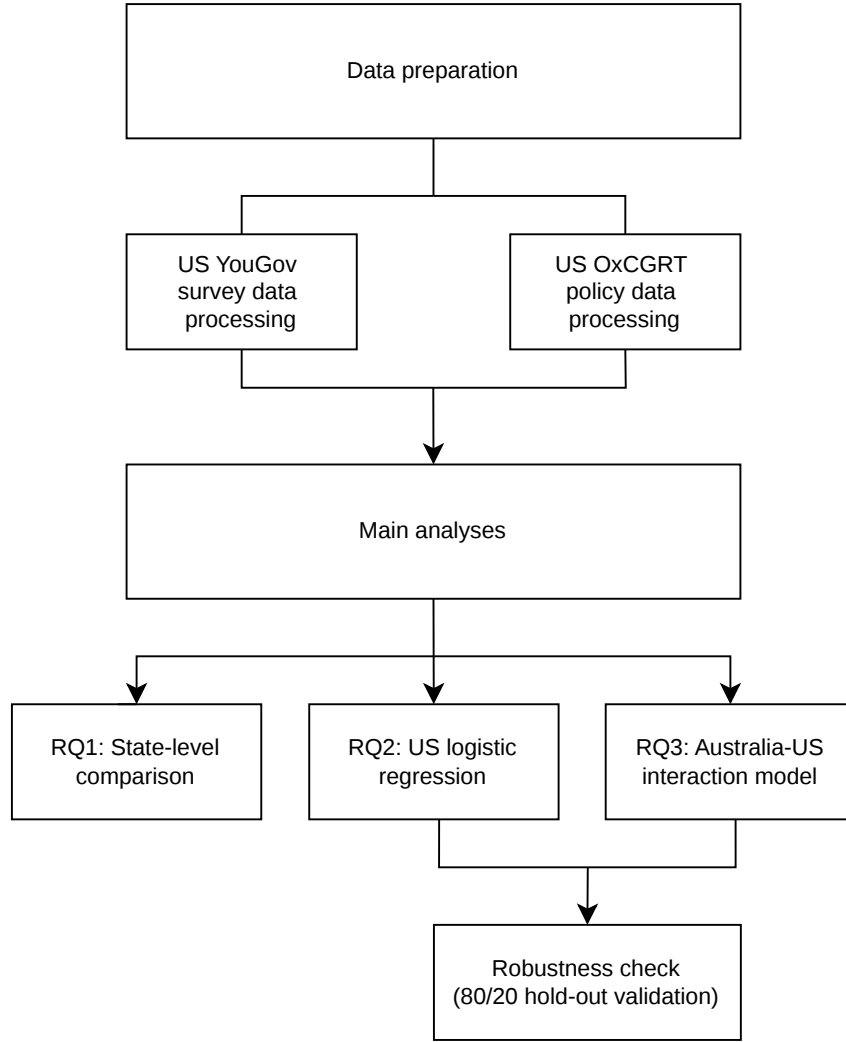


Figure 1: Analytical framework

policy, recommended wearing, required wearing in designated public places, required wearing in all public places or situations where social distancing cannot be maintained, and required wearing in all situations outside the home, respectively.

5.5 US YouGov data processing

5.5.1 Missing values

First, the raw survey data underwent a missing value check. After the raw data was read, missing values were identified for all variables. Empty strings, fields containing only spaces, and missing markers from the data source (such as “NA”) were all considered missing values (`0_missing_table_usa.py`, lines 14-20).

For each variable, the number of missing values and the missing percentage were calculated, where the missing percentage was defined as the proportion of missing observations to the total number of observations in the raw data. This step was only used to assess the data integrity of the variables and generate missing value statistics (`0_missing_table_usa.py`, lines 22-27).

Subsequently, following the variable screening logic in Ryan et al. (Ryan et al. 2025)’s Australian analysis, variables with a missing percentage exceeding 20% were initially removed; it means variables with more than 6,788 missing values were removed (`1_clean_dataset_usa.py`, lines 51-52). However, core variables relevant to the research question, model adjustment, and cross-country comparisons were retained for subsequent processing, even if their missing percentage exceeded this threshold (`1_clean_dataset_usa.py`, lines 54-83). The missing-value summary are recorded in Appendix B.1.

5.5.2 Face mask wearing

The mask-wearing behaviour variable was used to construct the primary outcome variable of this study. This variable was constructed based on four mask-wearing items from the YouGov survey (see Appendix B.2):

- wearing masks outside the home;
- wearing masks inside grocery stores/supermarkets;
- wearing masks inside clothing/footwear shops; and
- wearing masks on public transportation.

First, the responses to the four items were converted into a uniform five-point frequency scale, where: Always = 5; Frequently = 4; Sometimes = 3; Rarely = 2; Not at all = 1 (`1_clean_dataset_usa.py`, lines 95-101). For each observation, the median of valid responses to the four items was calculated, constructing a face mask-wearing frequency scale (lines 108-116). Because this study aimed to compare data from the US and Australia, the median used by Ryan et al. was adopted for calculation (line 118-121).

Then, following the standard set by Ryan et al. (Ryan et al. 2025), observations with a median of 4 or higher were defined as the high mask-wearing group and coded as 1; observations with a median of less than 4 were coded as 0 (lines 123-125).

5.5.3 Non-mask protective behaviour

To control for respondents’ general protective behaviours other than mask wearing, this study further constructed an indicator of non-mask protective behaviour.

First, the remaining protective behaviour items were coded using the same five-point frequency scale as the mask wearing item mentioned above (`1_clean_dataset_usa.py`, line 127-131). Then, the median of all valid protective behaviour items for each respondent was calculated to construct a protective behaviour indicator, `protective_behaviour_scale`, which represents the overall protective behaviour scale (line 133-140).

Since mask wearing behaviour is the primary outcome variable, including mask-related items in the protective behaviour indicator could lead to duplicate information between explanatory and outcome variables in the model. Therefore, when constructing the non-mask protective behaviour indicator for control in the regression model, this study removed the four mask

wearing items mentioned above and only used other protective behaviour items to calculate `protective_behaviour_nomask_scale`, representing the non-mask protective behaviour scale (line 142-150).

This indicator controls for individuals’ overall protective tendencies without considering mask use, such as reducing contact, isolation, or other preventative behaviours, thereby avoiding confusion between general protective awareness and mask wearing behaviour (The specific items and corresponding behavioural meanings are shown in Appendix B.3).

5.5.4 Covariates

This study selects control variables based on the research question and the model specification in Ryan et al. (Ryan et al. 2025)’s analysis of Australia, including demographic characteristics, health-related variables, government trust variables, and mental health variables.

The household size variable was converted to a numeric variable, with values for 1 to 7 family members retained. “8 or more” was coded as 8, while “Prefer not to say,” “Don’t know,” and missing responses were treated as missing values (`1_clean_dataset_usa.py`, lines 18-34).

The age variable was retained as a continuous numeric variable. Gender and employment status were treated as categorical variables (line 54-83).

5.5.5 Time index

To control for variations in the survey period, this study reconstructed the time variable instead of directly using the original survey week variable, `qweek`.

First, the earliest survey date in the US YouGov database was used as the starting point. Then, based on the time difference between the respondents’ survey date and the starting date, the entire survey period was divided into consecutive 14-day intervals. The first 14-day interval was coded as 1, and the code increased by 1 for each subsequent new 14-day interval (`1_clean_dataset_usa.py`, line 155-157).

5.5.6 Analytical sample

After variable cleaning and construction, the final sample for subsequent statistical analysis was determined.

Records were retained only when valid information was available for state, survey completion date, the mask-wearing scale, the binary mask-wearing outcome, the non-mask protective behaviour measure, gender, age, household size and employment status (`1_clean_dataset_usa.py`, line 159-174). Records lacking any of the above key variables were deleted.

Then, remove original variables that have been used to construct new variables and are no longer needed, including: original `qweek` time variables; original mask-wearing items; weights not used in subsequent models (line 176-1780). The sample-selection process is summarised in Appendix B.4.

The final cleaned US survey dataset is saved as `cleaned_data_usa.csv`, which is used for subsequent linking with the state-level sustained mask mandate start date and for further model preprocessing.

5.6 OxCGRT policy data processing

5.6.1 US policy data

US state-level policy data were extracted from OxCGRT compact subnational policy dataset (`000_filter_us_policy.py`, line 15). Because this study investigates the association between state-level mask mandate policies and individual mask-wearing behaviour in the US, the dataset was restricted to state-level policy records from the US (line 18). National-level records and policy observations from other jurisdictions were excluded.

The filtered dataset retained seven variables: country name, country code, region name, region code, jurisdiction level, policy date, and the H6M Facial Coverings indicator (line 23-32). Among them, the region name is used to distinguish different states, the policy date is used to establish the policy timeline for each state, and the H6M Facial Coverings indicator is used to indicate the strictness of the mask policy on that day (see Section 5.4.2). The processed US state-level policy data is saved as `OxCGRT_US_latest.csv`. A summary of `OxCGRT_US_latest.csv` in Appendix C.1.

5.6.2 Sustained mandate start date

First, this study read the file `OxCGRT_US_latest.csv`, converted the Date data to datetime format, and set a time index (`00_mask_mandates_usa.py`, line 14-21). Then, this study grouped data by state. The purpose of this was to ensure that changes to mask policies in each state were calculated separately within that state, rather than mixing policy records from different states together. After grouping by state, this study calculated a 14-day rolling average of the strictness of mask policies for each state (line 23-27).

The 14-day rolling average means that for each day, this study uses the policy strictness records of that day and the preceding 13 days to calculate the average level over those 14 days. The reason for using a 14-day rolling average is that daily policy records can be affected by short-term adjustments, temporary changes, or data fluctuations. Using only the policy value of a single day might misinterpret short-term or sporadic policy changes as sustained mask mandates. Therefore, the 14-day rolling average provides a more stable reflection of whether a state has truly entered a sustained mask policy phase.

After calculating the rolling average, 3 was set as the threshold for identifying sustained mask mandates (line 29-33). For each state, the date on which the 14-day rolling average first occurred greater than or equal to 3 was defined as the state's sustained mask mandate start date. This threshold follows the policy definition in Ryan et al.'s Australian analysis to ensure comparability of the results from the two analyses.

Finally, only the start date when each state first met the condition was retained and saved as `us_mandate_start_dates.csv`. See Appendix C.2 for the sustained mask mandate start dates identified by each state.

5.6.3 Mandate-period indicator

First, the sustained mask mandate start dates identified in the previous section 5.6.2 are matched against the individual survey data processed in 5.5 based on the respondents' reported state of

residence (`3_data_preprocessing_usa.py`, line 50-56). Before merging, the District of Columbia naming in both the survey and policy data is standardized to ensure correct matching (line 36-39).

The sustained mask mandate indicator was coded as follows (line 15-24):

- Survey date on or after the state’s mandate start date: 1;
- Survey date before the state’s mandate start date: 0; and
- No identified state mandate start date: 0.

After constructing the sustained mask mandate period indicator, the state name and start date fields used for data matching are deleted (line 76-79), retaining only the final constructed binary sustained mask mandate period indicator for subsequent state-level descriptive and regression analyses.

5.7 RQ1: State-level comparison

To describe the changes in mask-wearing rates before and after sustained mask mandates across different U.S. states, this study conducted state-level descriptive comparisons. This analysis was intended to demonstrate observed differences in mask-wearing behaviour between states and was not intended to estimate the causal effects of mask policies.

Step 1: Selection of indicators and state-period grouping. The binary high mask-wearing indicator constructed in Section 5.5.2 (`2_state_level_descriptive_comparison.py`, line 28-44) and the sustained mandate-period indicator constructed in Section 5.6.3 (line 51-64) were used in the state-level comparison analysis. Respondents were grouped according to their reported state and mandate-period status. For each state, respondents were grouped as either before the sustained mandate period or during the sustained mandate period. States without an identified sustained mandate start date had no during-mandate group (line 68-73).

Step 2: Calculation of state-period proportions. For each state-period group, the total number of respondents, the number classified as having high mask-wearing frequency, and the observed proportion with high mask-wearing frequency were calculated (line 82-134).

For state s in mandate-period category m , the observed proportion with high mask-wearing frequency was calculated as

$$\hat{P}_{s,m} = \frac{K_{s,m}}{N_{s,m}}, \quad (3)$$

where $\hat{P}_{s,m}$ denotes the observed high mask-wearing proportion in state s during mandate-period category m ; $K_{s,m}$ is the number of respondents classified as having high mask-wearing frequency in that state-period group; and $N_{s,m}$ is the total number of respondents in the same group. The period category m corresponds to the value of the individual mandate-period indicator: $m = 0$ denotes the pre-mandate period and $m = 1$ denotes the sustained mandate period for states with an identified onset date.

Step 3: Calculation of within-state proportional change. For each state with respondents before and during an identified sustained mandate period, the proportional change in high mask-wearing was calculated by comparing the during-mandate proportion with the corresponding pre-mandate proportion (line 136-151):

$$R_s = 100 \left(\frac{\hat{P}_{s,1} - \hat{P}_{s,0}}{\hat{P}_{s,0}} \right), \quad (4)$$

where R_s denotes the percentage change in the observed high mask-wearing proportion in state s ; $\hat{P}_{s,0}$ is the observed pre-mandate proportion; and $\hat{P}_{s,1}$ is the observed proportion during the sustained mandate period. Positive values of R_s indicate a higher observed proportion during the sustained mandate period, whereas negative values indicate a lower observed proportion.

States without an identified sustained mandate start date were excluded from the proportional-change figure because no during-mandate proportion could be calculated for these states (line 133-134). State-period sample sizes and observed proportions are reported in Appendix D.1.

5.8 RQ2: US logistic regression

To estimate the adjusted association between the period of the sustained mask mandate and high-frequency mask-wearing behaviour in the US sample, this study used a logistic regression model. The specific steps are as follows.

Step 1: Outcome coding. The mask-wearing behaviour variable constructed in Section 5.5.2 was recoded as a binary dependent variable. High mask-wearing behaviour was coded as 1, and all other cases were coded as 0 (`4_basic_logistic_usa.py`, lines 26–30).

Step 2: Exposure definition. The sustained mask mandate period indicator constructed in Section 5.6.3 was used as the main explanatory variable for the model, where 1 indicates that the respondents completed the survey during the sustained mask mandate period, and 0 indicates other circumstances (lines 32–43).

Step 3: Covariate preparation. Before modeling, all categorical variables were converted to dummy variables (lines 50–68). The categorical variables included: gender; employment status; previous health status; vaccination willingness; evaluation of the government response; confidence in government; and PHQ-4 items.

Each categorical variable retained a reference category, and the remaining categories were converted to binary indicator variables to avoid perfect multicollinearity. Numerical control variables included age; household size; number of children in the household; three health-related indicators; and protective behaviour without a mask (lines 32–43).

Step 4: Construction of the model-specific sample. Then, all variables are converted to numerical values (line 75-79). If any model variable has missing values, the corresponding observation is deleted (line 81-85).

The US model was specified as:

$$\text{logit} [\Pr (Y_i = 1)] = \beta_0 + \beta_M M_i + \beta_X^\top \mathbf{X}_i, \quad (5)$$

where Y_i denotes the binary high mask-wearing outcome for respondent i ; M_i denotes the respondent's sustained mandate-period status; $\mathbf{X}_i$ is the vector containing the included covariates and the fortnight time term; β_0 is the intercept; β_M is the log-odds coefficient for sustained mandate-period exposure; and β_X is the vector of coefficients corresponding to the included covariates.

Step 5: Model estimation and reporting. An intercept was included, and the logistic regression model was estimated by maximum likelihood. The numerical optimisation procedure was permitted a maximum of 200 iterations (line 87-92).

Step 6: Robustness check. To examine whether the main results of the US Logistic Regression model are relatively stable for sample partitioning, this study further conducted a hold-out robustness check. This robustness check used the same modeling data, dependent variable, principal explanatory variables, and control variables as the main model.

First, the full-case modeling sample was randomly divided into training and testing samples. This study used an 80/20 hold-out split, meaning 80% of the records were used as the training sample and 20% as the testing sample. The random seed was set to 20240417 to ensure reproducibility. Simultaneously, stratified sampling was used according to the dependent variable to make the proportion of high mask-wearing in the training and testing samples as close as possible. (4b_rq2_holdout_robustness_usa.py, line 111-120)

Then, this study fitted the same Logistic Regression model to the full sample, training sample, and testing sample (line 122-148). For each model, the regression coefficients, odds ratio, 95% confidence interval, and two-sided p-values were saved. Here, the testing sample is not only used for prediction, but is also fitted with a logistic regression separately to check whether the estimated direction, odds ratio, and statistical significance of the sustained mandate-period indicator are consistent across different samples.

5.9 RQ3: Australia-US harmonisation and interaction model

To examine whether the association between sustained mask mandate periods and high mask-wearing differed between Australia and the US, the data were harmonised before fitted the interaction model. The procedure comprised the following steps:

Step 1: Country identification. A country label was added to each respondent record before the two datasets were combined. Australian records were labelled as Australia and US records as US. (5_prepare_combined_aus_us.py, line 30-35)

Step 2: Outcome harmonisation. In both datasets, respondents classified as having high mask-wearing frequency were coded 1, and respondents in the comparison category were coded 0 (line 37-49). This produced a common binary outcome, denoted by Y_i , for respondent i .

Step 3: Mandate-period harmonisation. The individual-level sustained mandate-period indicator was converted to numeric form in both datasets. Respondents recorded during a sustained mandate period were coded 1, others were coded 0. (line 51-63)

Step 4: Selection of common numeric covariates. Only numeric variables available in both country datasets were eligible for inclusion in the harmonised dataset. The common numeric covariates were age, household size, health-related variables and the non-mask protective behaviour measure. Variables not available in both datasets were removed. (line 65-87)

Step 5: Selection of comparable categorical indicators. The binary indicators in both datasets include gender, employment status, previous health status, vaccination willingness, evaluation of the government response, confidence in government, and the PHQ-4 items. However, only indicator columns representing the same response category and present in both datasets were retained. Then, country-specific indicator categories not present in both dataset were excluded (line 92-127). The complete mapping is shown in Appendix F.

Step 6: Dataset combination. Then the Australian and US data were restricted to the same variable columns, merged by row, and combined into a single dataset for individual-level joint analysis (line 129-142).

Step 7: Country and interaction coding. After that a binary country indicator, G_i , was coded 0 for Australia and 1 for the US (line 144-152). The product $M_i G_i$ was then constructed to represent the interaction between mandate-period exposure and country.

The country indicator was defined as

$$G_i = \begin{cases} 0, & \text{if respondent } i \text{ was from Australia,} \\ 1, & \text{if respondent } i \text{ was from the United States.} \end{cases} \quad (6)$$

The country-by-mandate interaction term was constructed as $M_i G_i$, where M_i denotes the individual-level sustained mandate-period indicator. It is coded 1 if respondent i 's survey record occurred during a sustained mandate period, and 0 otherwise.

Step 8: Core Variable Completeness Screening. Convert the four core variables into numerical variables, including outcome variables, policy period variables, country variables, and interaction terms. If missing values exist in these variables, remove them from the merged dataset (line 155-170).

Step 9: Interaction logistic regression model. After constructing the harmonised Australia-US dataset, a binary logistic regression model was fitted.

The model was specified as

$$\text{logit} [\Pr (Y_i = 1)] = \theta_0 + \theta_M M_i + \theta_G G_i + \theta_{MG} (M_i G_i) + \boldsymbol{\theta}_X^T \mathbf{X}_i, \quad (7)$$

where Y_i denotes the binary high mask-wearing outcome for respondent i ; M_i is the sustained mandate-period indicator; G_i is the country indicator; $M_i G_i$ is the country-by-mandate interaction; $\mathbf{X}_i$ is the vector of common covariates; θ_0 is the intercept; θ_M is the mandate-period log-odds coefficient for Australia; θ_G is the US-Australia log-odds difference when $M_i = 0$; θ_{MG} is the difference between the US and Australian mandate-period log-odds associations; and $\boldsymbol{\theta}_X$ is the vector of covariate coefficients.

Step 10: Model estimation and reporting. An intercept was included, and the logistic regression model was estimated by maximum likelihood. The numerical optimisation procedure was permitted a maximum of 300 iterations. (`6_country_interaction_model.py`, line 89-109)

Step 11: Robustness check. To verify whether the results of the Australia-US interaction model are stable for sample partitioning, this study further performed a hold-out robustness check. This robustness check used the same harmonized Australia-US dataset as the main interaction model and maintained the same model variable structure.

First, the full modeling samples were randomly divided into training and testing samples. This study uses an 80/20 hold-out split, meaning 80% of the records are used as training samples and 20% as testing samples. The random seed was set to 20240417 to ensure the same sample partitioning for each run. The sample partitioning was stratified by “country + high mask-wearing outcome,” ensuring that the proportions of Australia/US and high mask-wearing outcomes in the training and testing samples remained similar. (6b_rq3_holdout_robustness_interaction.py, line 131-147)

Then, this study fitted the same configured interaction logistic regression model to the full sample, training sample, and testing sample. (line 149-174) The regression coefficients, odds ratios, 95% confidence intervals, and two-sided p-values of all variables are preserved.

6 Results

6.1 Data preparation

6.1.1 US Policy data and mandate start date

After applying the policy data processing and mandate-date construction described in Section 5.6, the filtered OxCGRT subnational policy data contained 55,896 US state-day records covering all 50 states and the District of Columbia. The retained policy data contained no missing H6M values. Additional information on the temporal coverage and completeness of the policy data is provided in Appendix C.1.

Of the 51 US states and jurisdictions included in the policy data, 47 reached the prespecified sustained mandate threshold. No sustained mandate onset date was identified for Minnesota, Nebraska, South Dakota or Wyoming because their 14-day rolling mean policy stringency did not reach the threshold of 3. Among the remaining jurisdictions, the identified onset dates ranged from 19 April 2020 to 13 September 2021. State-specific onset dates are reported in Appendix C.2.

6.1.2 US YouGov survey sample

After applying the data processing and measure construction steps described in Section 5.5, the original US survey dataset contained 33,940 records and 424 variables. After missing value filtering, variable selection, and complete case filtering, the final US survey sample included 32,458 respondents from all 50 states and the District of Columbia. The survey completion dates ranged from April 2, 2020, to October 15, 2021. Of the cleaned sample, 20,477 respondents (63.09%) were categorized as high mask-wearing, while the remaining 11,981 (36.91%) were in the comparison category. A summary of the cleaning steps is provided in Appendix B.5.

6.2 RQ1: State-level comparison

The state-level descriptive comparison showed (Figure 2) that among the 47 states with an identified sustained mandate start date, 33 had a higher observed proportion of high mask-wearing

during the sustained mandate period than before its start date, while 14 had a lower observed proportion. State-level proportional changes ranged from -11.95% in Colorado to 225.00% . Montana, Minnesota, Nebraska, South Dakota and Wyoming had no identified sustained mandate start date. Complete state-period sample sizes and proportions are reported in Appendices C.1 and C.2.

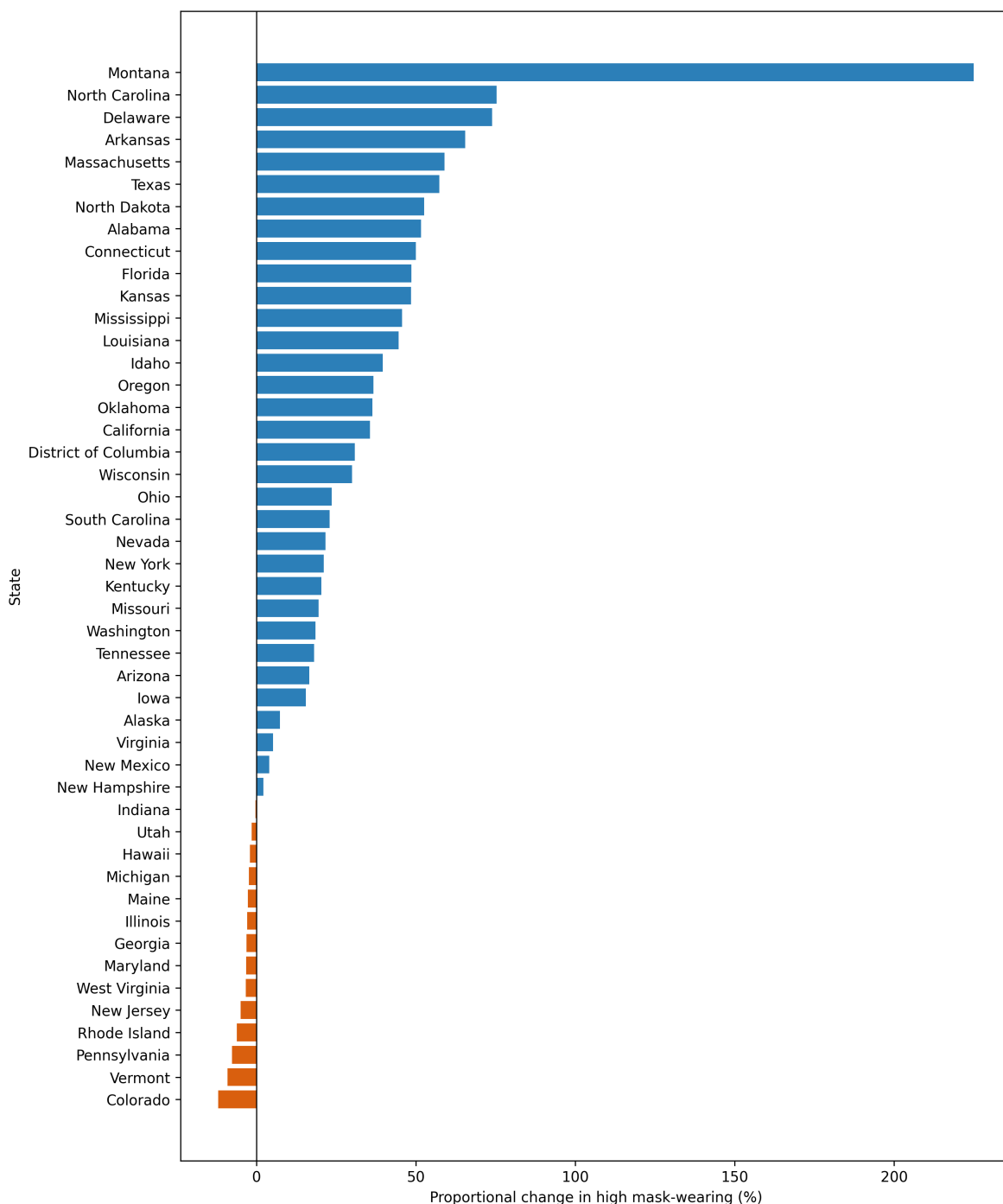


Figure 2: State-level proportional change in high mask-wearing during sustained mandate periods.

6.3 RQ2: US logistic regression

The key estimate is reported in Table 1, and the complete model estimates are reported in Appendix E.

Table 1: Key estimate of the US logistic regression.

Effect	Model sample	Adjusted odds ratio	95% confidence interval	p-value
Sustained mandate-period association	26,176	1.684	1.542 to 1.839	<0.001

6.3.1 US model robustness check

The US model sample was divided into 20,940 training observations and 5,236 testing observations (see Appendix G.1). The adjusted mandate-period odds ratio can be found in Figure 3. Detailed coefficient estimates are reported in Appendix G.2.

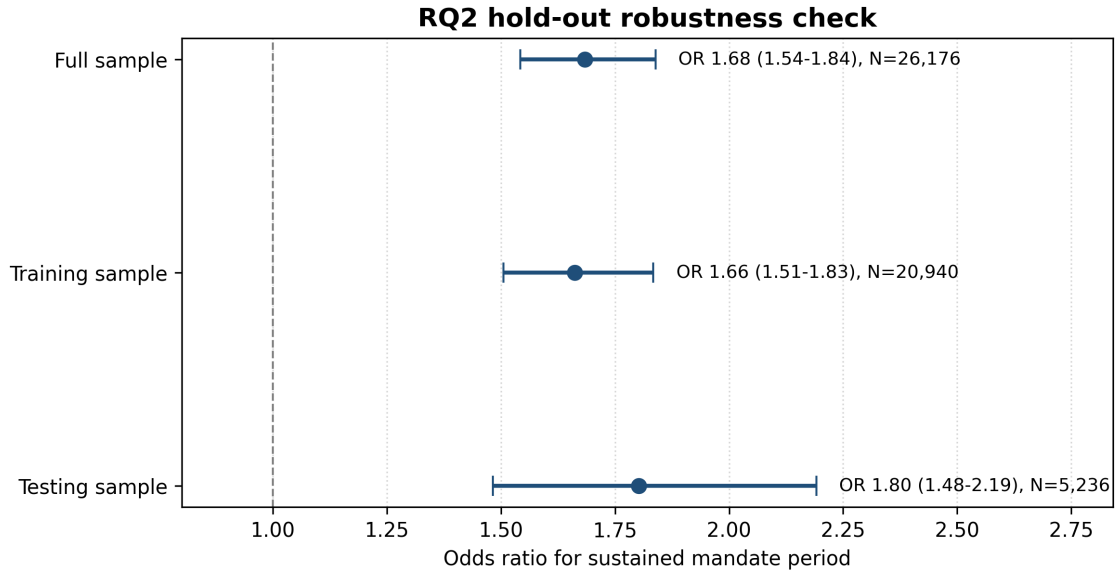


Figure 3: Mandate-period odds ratios in the full, training and testing US samples.

6.4 RQ3: Australia-US harmonisation and interaction model

The harmonised Australia-US dataset contained 72,594 respondent records: 40,136 from Australia and 32,458 from the US (see Appendix F.2). In this dataset, high mask-wearing was observed in 53.96% of Australian records and 63.09% of US records. Sustained mandate-period observations accounted for 62.76% of Australian records and 68.69% of US records.

With Australia as the reference country, the key estimates are reported in Table 2.

Table 2: Key estimates of the Australia–US interaction model.

Term	Adjusted odds ratio	95% confidence interval	p-value
Mandate period: Australia	6.971	6.574 to 7.393	<0.001
Country: US vs Australia at mandate = 0	4.846	4.496 to 5.224	<0.001
Country-by-mandate interaction	0.211	0.194 to 0.229	<0.001

6.4.1 Interaction model robustness check

The full interaction-model sample contained 72,594 respondents, of whom 58,075 were allocated to the training sample and 14,519 to the testing sample (see Appendix G.1). The adjusted country-by-mandate interaction odds ratio can be find in Figure 4. Detailed coefficient estimates are reported in Appendix G.3.

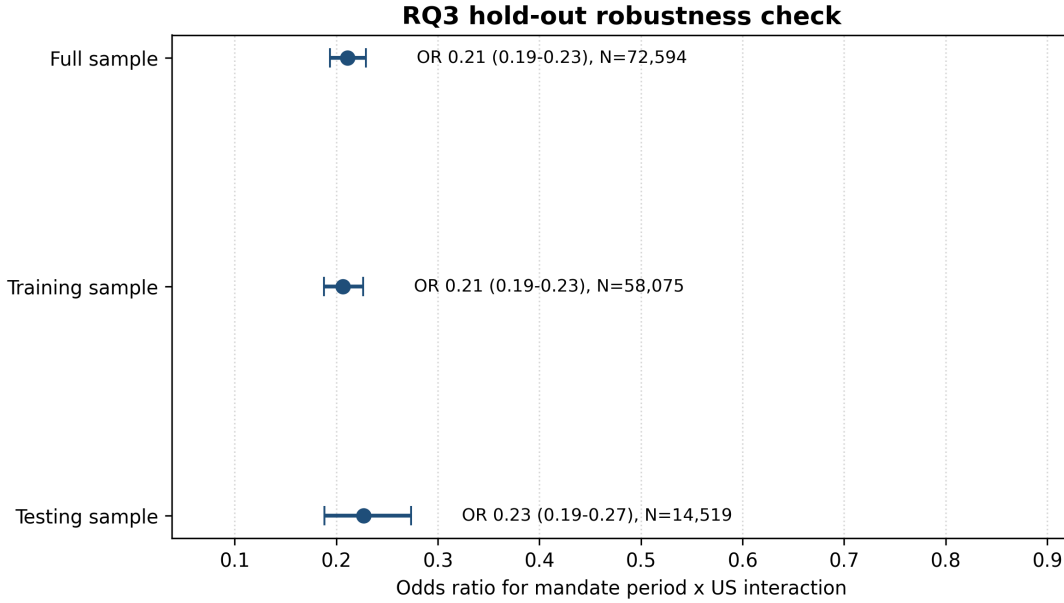


Figure 4: Country-by-mandate interaction estimates in the full, training and testing samples.

7 Discussion

7.1 Main findings

The following are the main findings corresponding to the three research questions in this study:

7.1.1 RQ1: State-level comparison

The results suggest that high mask-wearing rates generally increased during sustained mandate periods in most US states and the state-level proportional change ranged widely. This indicates that there is significant state-level heterogeneity and inconsistency in changes in mask-wearing before and after the mandate period across different US states.

7.1.2 RQ2: US logistic regression

The results showed that even after controlling for confounding factors such as demographic characteristics, health status, other protective behaviours and survey duration, a positive association remains between the mandate period and higher frequency of mask-wearing. This indicates that even after considering multiple factors that may influence residents' behaviour, the sustained mask mandate remains stably associated with higher levels of mask-wearing behaviour.

Robustness tests showed that the association direction remained consistent across the complete sample, training sample, and test sample, indicating that this positive correlation was not caused by a single sample division. However, this study is an observational study and can only demonstrate a statistical association between the period of mandatory mask-wearing and high-frequency mask-wearing behaviour, but cannot infer a direct causal relationship between the two.

7.1.3 RQ3: Australia-US interaction model

The results show that this association is positive in both countries, but the association is much weaker in the US than in Australia. This result suggests that the policy-behaviour relationship in the Australian replication framework cannot be simply generalized to the US, because there are significant differences in the policy environment and behavioural response patterns between the two countries. This difference may reflect variation in policy implementation, state-level governance, public attitudes, social norms and behavioural responses between the two countries.

The robustness checks showed a consistent direction across the full, training and testing samples, which supports the stability of this overall pattern. However, this result should still be interpreted as a difference in cross-national statistical association, rather than a strict causal effect.

7.2 Comparison with existing literature

These findings are consistent with existing literature. The state-level differences found in RQ1 are consistent with evidence that US mask-wearing varied across time and space (Taube, Susswein, and Bansal 2023, 1–2). The RQ2 finding that sustained mask mandate periods were associated with higher mask-wearing rates is consistent with previous evidence (Näher et al. 2024, 6; MacIntyre et al. 2021, 205; Haischer et al. 2020, 7–8). The differences between Australia and the US in RQ3 are consistent with evidence that mask acceptance varies with cultural and socio-cultural contexts, including collectivism, social norms, policy signals, and local policy environments (Lu, Jin, and English 2021, 1; Yang et al. 2022, 1–2).

7.3 Study contribution

The main contribution of this study is that it extends the research to the US based on the Australian analytical framework of Ryan et al., uses a relatively transparent and interpretable method, and finally uses a robustness check to test the reliability and stability of the results.

7.4 Limitations and future directions

7.4.1 Self-reported behaviour

The high mask-wearing outcome in this study was derived from respondents' self-reports in the YouGov COVID-19 Behaviour Tracker, recall bias and social desirability bias may exist (Hutchins et al. 2020, 1587). During the pandemic, mask-wearing is not only a personal protective behaviour but may also be influenced by social norms and politics. Therefore, when comparing different states or countries, different regional social norms may influence the way self-reporting is conducted.

Future research could use more diverse data sources, such as mobile behaviour data, observational public space data, or other data that indirectly reflect actual mask-wearing behaviour. This can reduce recall bias and social desirability bias arising from relying solely on self-reporting.

7.4.2 Median-based outcome

To maintain consistency with the Australian analysis conducted by Ryan et al., the primary dependent variable of high mask-wearing was constructed by calculating the median of each respondent's valid responses to four questions, defining respondents with a median of 4 or higher as the high mask-wearing group. (See Appendix B.2 for the specific four questions.) However, the median is insensitive to extreme values (Mishra et al. 2019, 68), which may therefore overlook differences in mask-wearing behaviour across different scenarios.

Future research could compare different outcome construction methods. For example, median-based scales, mean-based scales, and total scores can be used simultaneously to define high mask-wearing. If different construction methods yield similar conclusions, the results will be more convincing; if the results differ significantly, further investigation can be conducted to explore how the measurement method of mask-wearing behaviour affects the model results.

7.4.3 Association, not causation

The model results in this study should be interpreted as statistical associations, not causal effects. Although the models controlled for covariates such as demographics, household structure, health-related variables, non-mask protective behaviour, and survey duration, this study used observational cross-sectional survey data and employed conventional logistic regression and Australia-US interaction logistic regression. Therefore, the results cannot demonstrate that sustained mask mandate periods directly caused higher mask-wearing behaviour (Hammerton and Munafò 2021, 563–66).

Future research could use stronger causal identification methods. For example, state fixed effects can be incorporated to control for stable differences between states, or difference-in-differences methods can be used to compare changes before and after policy implementation. This would allow for a more rigorous discussion of the causal impact of mask mandate on mask-wearing behaviour.

7.4.4 Limited Country Comparison

This study’s country-level comparison included only Australia and the US, so the generalisability of the findings to other countries is limited (Franke and Richey 2010, 1275). Although the Australia-US interaction model can compare the strength of the association between sustained mandate period and high mask-wearing between the two countries, this comparison is still limited by the limited number of countries.

Future research could extend the comparison to additional countries or regions, thereby testing whether the findings observed in Australia and the US also apply in broader international contexts (Franke and Richey 2010, 1284).

8 Conclusion

This study investigates the association between sustained mask mandate periods and high mask-wearing behavior. It first conducts descriptive comparisons between US states, then estimates the association using a US logistic regression model, and finally compares whether this association differs between the two countries using an Australia-US interaction model. Overall, the results show a positive association between sustained mandate periods and higher mask-wearing behavior, but this association varies significantly across different US states and its strength differs between the US and Australia.

9 Acknowledgements

I would like to thank the teaching team and classmates for their help, feedback, and support throughout the project. Their suggestions helped me further clarify the research question, improve the analytical structure, and better understand how to interpret the research methods and results.

I also acknowledge that I used AI tools to assist in language polishing, expression modification, and code understanding during the preparation of this study. However, the research design, data processing, model analysis, results interpretation, and final writing decisions were all completed and reviewed by myself.

All code used in this study can be found in the OneDrive folder **AnthonyMays-Covid_Ma_J/PartB_US_Extension/code**. The full script name will be used the first time the code is cited in the text; if the same script is cited again later, only the relevant line number will be used to avoid duplication and save words.

All code, data-processing scripts, and supporting materials used to reproduce the analyses and results in this report are available at: https://github.com/mjyhub/MDSresearch_FinalReport_Ma_J

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11 Statistical appendix

11.1 Appendix A: Key statistical terms

Table 3: Key statistical terms used in model interpretation.

Term	Formula	Interpretation
Odds	$p/(1-p)$	The relative probability of an event occurring rather than not occurring, where p is the probability of the event. Larger odds indicate that the event is more likely to occur.
Odds ratio (OR)	$\exp(\beta)$	The relative change in odds associated with a predictor, where β is the logistic regression coefficient. $OR > 1$ indicates higher odds, $OR < 1$ indicates lower odds, and $OR = 1$ indicates no difference in odds.
95% confidence interval	$[L, U]$	The uncertainty range around an estimate. For an OR, if the interval includes 1, the evidence for an association is weaker. If the interval is entirely above or below 1, the association is more statistically supported.
p -value	Reported by model	Evidence against the null hypothesis. Smaller values provide stronger evidence against no association.
Interaction term	$A \times B$	Tests whether the association between A and the outcome changes depending on B . If the interaction OR is close to 1, it indicates a small difference in association between the two groups; if interaction $OR > 1$, it indicates a stronger association in the target group compared to the reference group; if interaction $OR < 1$, it indicates a weaker association in the target group.

11.2 Appendix B: US YouGov data processing

11.2.1 Appendix B.1: Missing-value summary

Source: Generated from `cleaned_data_usa.csv`, produced by `1_clean_dataset_usa.py`.

Table 4: Distribution of variable-level missingness in the original US survey data.

Missingness category	Number of variables
0%	25
>0% to 10%	2
>10% to 20%	4
>20% to 50%	40
>50% to 80%	46
>80%	307

Table 5: Core variables retained despite exceeding the 20% missingness threshold.

Source variable	Missing observations	Missing percentage
i10_health	6,987	20.59%
WCRex1	7,950	23.42%
i12_health_22	9,947	29.31%
i12_health_23	9,947	29.31%
i12_health_25	9,947	29.31%
PHQ4_1	14,026	41.33%
PHQ4_3	14,044	41.38%
PHQ4_4	14,047	41.39%
PHQ4_2	14,057	41.42%

11.2.2 Appendix B.2: Mask-wearing outcome items

Table 6: Items used to construct the high mask-wearing outcome.

Setting	Source variable	Response coding
Outside the home	i12_health_1	1 = Not at all; 2 = Rarely; 3 = Sometimes; 4 = Frequently; 5 = Always
Grocery store or supermarket	i12_health_22	1 = Not at all; 2 = Rarely; 3 = Sometimes; 4 = Frequently; 5 = Always
Clothing or footwear shop	i12_health_23	1 = Not at all; 2 = Rarely; 3 = Sometimes; 4 = Frequently; 5 = Always
Public transportation	i12_health_25	1 = Not at all; 2 = Rarely; 3 = Sometimes; 4 = Frequently; 5 = Always

11.2.3 Appendix B.3: Non-mask protective behaviour items

Source: Generated from `cleaned_data_usa.csv`, produced by `1_clean_dataset_usa.py`.

Table 7: Items used to construct the non-mask protective behaviour measure.

Source variable	Behaviour represented
i12_health_2	Washing hands with soap and water
i12_health_3	Using hand sanitiser
i12_health_4	Covering the nose and mouth when coughing or sneezing
i12_health_5	Avoiding contact with symptomatic or potentially exposed people
i12_health_6	Avoiding going out

Source variable	Behaviour represented
i12_health_7	Avoiding hospitals or other healthcare settings
i12_health_8	Avoiding public transportation
i12_health_11	Avoiding having guests at home
i12_health_12	Avoiding small social gatherings
i12_health_13	Avoiding medium-sized social gatherings
i12_health_14	Avoiding large social gatherings
i12_health_15	Avoiding crowded areas
i12_health_16	Avoiding shops

11.2.4 Appendix B.4: Sample-selection details

Source: Based on missing-value assessment from `missing_value_counts_usa.csv` produced by `0_missing_table_usa.py`, and `cleaned_data_usa.csv` produced by `1_clean_dataset_usa.py`. The complete-case logistic regression sample was generated by `03_basic_logistic_usa.py`.

Table 8: Construction of the cleaned US survey dataset and logistic regression sample.

Sample stage	Records retained	Records excluded at stage
Original US survey data	33,940	Not applicable
Cleaned US survey dataset	32,458	1,482
Complete-case US logistic regression	26,176	6,282

11.2.5 Appendix B.5: US Survey Data Cleaning and Analytical Sample Summary

Source: Generated from `united-states.csv`, `missing_value_counts_usa.csv`, and `cleaned_data_usa.csv`; intermediate outputs were produced by `0_missing_table_usa.py` and `1_clean_dataset_usa.py`.

Table 9: Summary of US survey data cleaning and final analytical sample.

Cleaning step / sample feature	Result
Raw US survey records	33,940
Raw variables	424
Variables with more than 20% missingness	393
Non-core high-missingness variables excluded	384
Core high-missingness variables retained	9
Records excluded by complete-case filtering	1,482
Final cleaned US sample	32,458
US regions covered	51 regions
Survey date range	02 Apr 2020 to 15 Oct 2021

Cleaning step / sample feature	Result
High mask-wearing respondents	20,477 (63.09%)
Comparison category respondents	11,981 (36.91%)

11.3 Appendix C: OxCGRT policy data processing

11.3.1 Appendix C.1: Summary of US state-level OxCGRT policy data

Source: Generated from `OxCGRT_US_latest.csv`, produced by `000_filter_us_policy.py`.

Table 10: US state-level policy data summary.

Characteristic	Value
Level of analysis	US state-day
Number of records	55,896
Number of regions	51
Date range	01 Jan 2020–31 Dec 2022
Missing H6M records	0

11.3.2 Appendix C.2: Sustained mandate start dates

Source: Generated from `us_mandate_start_dates.csv`, produced by `00_mask_mandates_usa.py`.

Table 11: State-specific sustained mandate start dates.

State	Sustained mandate start date
California	19 Apr 2020
Florida	26 Apr 2020
Texas	26 Apr 2020
Massachusetts	30 Apr 2020
New York	30 Apr 2020
Connecticut	03 May 2020
North Carolina	03 May 2020
Delaware	11 May 2020
Alabama	14 May 2020
Rhode Island	21 May 2020
Washington DC	26 May 2020
New Mexico	28 May 2020
Louisiana	29 May 2020
Mississippi	17 Jun 2020
Arkansas	29 Jun 2020

Arizona	02 Jul 2020
Nevada	07 Jul 2020
Oregon	07 Jul 2020
Pennsylvania	07 Jul 2020
Montana	09 Jul 2020
South Carolina	09 Jul 2020
Washington	09 Jul 2020
Utah	10 Jul 2020
Alaska	12 Jul 2020
Missouri	14 Jul 2020
Hawaii	15 Jul 2020
Kansas	16 Jul 2020
Tennessee	16 Jul 2020
Oklahoma	20 Jul 2020
New Jersey	21 Jul 2020
Ohio	21 Jul 2020
Kentucky	23 Jul 2020
Idaho	27 Jul 2020
Colorado	29 Jul 2020
Wisconsin	29 Jul 2020
Michigan	30 Jul 2020
Iowa	03 Aug 2020
Indiana	09 Aug 2020
Maryland	13 Aug 2020
Vermont	14 Aug 2020
Illinois	07 Oct 2020
North Dakota	01 Nov 2020
Maine	17 Nov 2020
West Virginia	26 Nov 2020
Virginia	29 Nov 2020
New Hampshire	03 Dec 2020
Georgia	13 Sep 2021
Minnesota	Not identified
Nebraska	Not identified
South Dakota	Not identified
Wyoming	Not identified

11.3.3 Appendix C.3: Individual-level policy matching

Source: Generated from `cleaned_data_preprocessing_usa.csv`.

Table 12: Individual-level mandate-period classification.

Mandate-period classification	Records	Percentage
Before/no sustained mandate period	10,164	31.31%
During sustained mandate period	22,294	68.69%

11.4 Appendix D: State-level descriptive results

11.4.1 Appendix D.1: State-period sample sizes and proportions

Source: Generated from `cleaned_data_usa.csv` during the state-level descriptive comparison, produced by `2_state_level_descriptive_comparison.py`.

Table 13: State-period sample sizes and observed high mask-wearing proportions.

State	Period	Sample size	High mask-wearing count	High mask-wearing proportion
Alabama	Before/no sustained mandate	117	47	40.2%
Alabama	During sustained mandate	427	260	60.9%
Alaska	Before/no sustained mandate	37	19	51.4%
Alaska	During sustained mandate	78	43	55.1%
Arizona	Before/no sustained mandate	254	117	46.1%
Arizona	During sustained mandate	488	262	53.7%
Arkansas	Before/no sustained mandate	91	35	38.5%
Arkansas	During sustained mandate	198	126	63.6%
California	Before/no sustained mandate	376	203	54.0%
California	During sustained mandate	2791	2043	73.2%
Colorado	Before/no sustained mandate	209	142	67.9%
Colorado	During sustained mandate	346	207	59.8%
Connecticut	Before/no sustained mandate	57	27	47.4%

State	Period	Sample size	High mask-wearing count	High mask-wearing proportion
Connecticut	During sustained mandate	273	194	71.1%
Delaware	Before/no sustained mandate	21	9	42.9%
Delaware	During sustained mandate	106	79	74.5%
District of Columbia	Before/no sustained mandate	32	21	65.6%
District of Columbia	During sustained mandate	85	73	85.9%
Florida	Before/no sustained mandate	348	159	45.7%
Florida	During sustained mandate	2080	1412	67.9%
Georgia	Before/no sustained mandate	1009	675	66.9%
Georgia	During sustained mandate	71	46	64.8%
Hawaii	Before/no sustained mandate	37	30	81.1%
Hawaii	During sustained mandate	68	54	79.4%
Idaho	Before/no sustained mandate	81	29	35.8%
Idaho	During sustained mandate	106	53	50.0%
Illinois	Before/no sustained mandate	609	418	68.6%
Illinois	During sustained mandate	656	437	66.6%
Indiana	Before/no sustained mandate	259	155	59.8%
Indiana	During sustained mandate	389	232	59.6%
Iowa	Before/no sustained mandate	93	44	47.3%
Iowa	During sustained mandate	225	123	54.7%
Kansas	Before/no sustained mandate	83	33	39.8%
Kansas	During sustained mandate	127	75	59.1%

State	Period	Sample size	High mask-wearing count	High mask-wearing proportion
Kentucky	Before/no sustained mandate	194	97	50.0%
Kentucky	During sustained mandate	344	207	60.2%
Louisiana	Before/no sustained mandate	66	32	48.5%
Louisiana	During sustained mandate	271	190	70.1%
Maine	Before/no sustained mandate	83	51	61.4%
Maine	During sustained mandate	87	52	59.8%
Maryland	Before/no sustained mandate	210	156	74.3%
Maryland	During sustained mandate	285	205	71.9%
Massachusetts	Before/no sustained mandate	90	41	45.6%
Massachusetts	During sustained mandate	533	386	72.4%
Michigan	Before/no sustained mandate	324	200	61.7%
Michigan	During sustained mandate	529	319	60.3%
Minnesota	Before/no sustained mandate	555	268	48.3%
Mississippi	Before/no sustained mandate	59	27	45.8%
Mississippi	During sustained mandate	189	126	66.7%
Missouri	Before/no sustained mandate	206	99	48.1%
Missouri	During sustained mandate	423	243	57.4%
Montana	Before/no sustained mandate	39	6	15.4%
Montana	During sustained mandate	98	49	50.0%
Nebraska	Before/no sustained mandate	175	97	55.4%
Nevada	Before/no sustained mandate	119	68	57.1%

State	Period	Sample size	High mask-wearing count	High mask-wearing proportion
Nevada	During sustained mandate	266	185	69.5%
New Hampshire	Before/no sustained mandate	82	49	59.8%
New Hampshire	During sustained mandate	95	58	61.1%
New Jersey	Before/no sustained mandate	370	274	74.1%
New Jersey	During sustained mandate	608	428	70.4%
New Mexico	Before/no sustained mandate	50	27	54.0%
New Mexico	During sustained mandate	121	68	56.2%
New York	Before/no sustained mandate	338	206	60.9%
New York	During sustained mandate	2271	1676	73.8%
North Carolina	Before/no sustained mandate	161	63	39.1%
North Carolina	During sustained mandate	745	511	68.6%
North Dakota	Before/no sustained mandate	25	8	32.0%
North Dakota	During sustained mandate	43	21	48.8%
Ohio	Before/no sustained mandate	438	214	48.9%
Ohio	During sustained mandate	795	480	60.4%
Oklahoma	Before/no sustained mandate	104	40	38.5%
Oklahoma	During sustained mandate	223	117	52.5%
Oregon	Before/no sustained mandate	167	84	50.3%
Oregon	During sustained mandate	333	229	68.8%
Pennsylvania	Before/no sustained mandate	544	366	67.3%
Pennsylvania	During sustained mandate	1054	655	62.1%

State	Period	Sample size	High mask-wearing count	High mask-wearing proportion
Rhode Island	Before/no sustained mandate	21	15	71.4%
Rhode Island	During sustained mandate	91	61	67.0%
South Carolina	Before/no sustained mandate	174	87	50.0%
South Carolina	During sustained mandate	322	198	61.5%
South Dakota	Before/no sustained mandate	74	27	36.5%
Tennessee	Before/no sustained mandate	230	108	47.0%
Tennessee	During sustained mandate	431	239	55.5%
Texas	Before/no sustained mandate	352	149	42.3%
Texas	During sustained mandate	1893	1261	66.6%
Utah	Before/no sustained mandate	81	46	56.8%
Utah	During sustained mandate	177	99	55.9%
Vermont	Before/no sustained mandate	30	22	73.3%
Vermont	During sustained mandate	57	38	66.7%
Virginia	Before/no sustained mandate	458	291	63.5%
Virginia	During sustained mandate	507	339	66.9%
Washington	Before/no sustained mandate	223	125	56.1%
Washington	During sustained mandate	486	323	66.5%
West Virginia	Before/no sustained mandate	109	67	61.5%
West Virginia	During sustained mandate	101	60	59.4%
Wisconsin	Before/no sustained mandate	233	103	44.2%
Wisconsin	During sustained mandate	402	231	57.5%

State	Period	Sample size	High mask-wearing count	High mask-wearing proportion
Wyoming	Before/no sustained mandate	67	28	41.8%

11.5 Appendix E: complete estimates of US logistic regression

Source: Generated from `cleaned_data_preprocessing_usa.csv` using `4_basic_logistic_usa.py`.

Table 14: Adjusted US logistic regression estimates.

Variable	Coefficient	Odds ratio	95% confidence interval	p-value
const	-3.336	0.036	0.027 to 0.047	<0.001
within_mandate_period	0.521	1.684	1.542 to 1.839	<0.001
age	0.001	1.001	0.999 to 1.004	0.256
household_size	-0.037	0.963	0.939 to 0.988	0.004
household_children	0.090	1.094	1.056 to 1.134	<0.001
i2_health	-0.000	1.000	1.000 to 1.000	0.876
i7a_health	-0.008	0.992	0.984 to 0.999	0.025
i13_health	-0.000	1.000	0.999 to 1.001	0.491
protective_behaviour_nomask_scale	0.680	1.974	1.921 to 2.028	<0.001
week_number	-0.020	0.980	0.976 to 0.985	<0.001
gender_Male	-0.258	0.773	0.727 to 0.821	<0.001
i9_health_No	-0.149	0.862	0.000 to Inf	1.000
i9_health_Not sure	-0.406	0.667	0.000 to Inf	1.000
i9_health_Yes	-0.249	0.779	0.000 to Inf	1.000
employment_status_Full time student	0.207	1.230	1.061 to 1.424	0.006
employment_status_Not working	-0.323	0.724	0.657 to 0.798	<0.001
employment_status_Other	-0.085	0.919	0.756 to 1.117	0.395
employment_status_Part time employment	0.013	1.013	0.911 to 1.126	0.813
employment_status_Retired	-0.328	0.721	0.654 to 0.794	<0.001
employment_status_Unemployed	-0.169	0.844	0.762 to 0.935	0.001
i11_health_Neither willing nor unwilling	-0.315	0.730	0.000 to Inf	1.000
i11_health_Not sure	-0.041	0.959	0.000 to Inf	1.000
i11_health_Somewhat unwilling	-0.329	0.720	0.000 to Inf	1.000
i11_health_Somewhat willing	0.086	1.090	0.000 to Inf	1.000
i11_health_Very unwilling	-0.511	0.600	0.000 to Inf	1.000
i11_health_Very willing	0.306	1.359	0.000 to Inf	1.000
WCRex1_Don't know	0.402	1.494	1.224 to 1.824	<0.001

Variable	Coefficient	Odds ratio	95% confidence interval	p-value
WCRex1_Somewhat badly	0.461	1.586	1.381 to 1.821	<0.001
WCRex1_Somewhat well	0.578	1.783	1.565 to 2.030	<0.001
WCRex1_Very badly	0.535	1.708	1.498 to 1.948	<0.001
WCRex1_Very well	0.522	1.686	1.452 to 1.958	<0.001
WCRex2_A fair amount of confidence	1.424	4.154	3.648 to 4.729	<0.001
WCRex2_A lot of confidence	1.688	5.408	4.657 to 6.281	<0.001
WCRex2_Don't know	1.357	3.883	3.181 to 4.741	<0.001
WCRex2_No confidence at all	1.210	3.352	2.860 to 3.929	<0.001
WCRex2_Not very much confidence	1.397	4.045	3.521 to 4.646	<0.001
PHQ4_1_More than half the days	-0.122	0.885	0.443 to 1.766	0.728
PHQ4_1_Nearly every day	-0.060	0.941	0.468 to 1.892	0.865
PHQ4_1_Not at all	0.140	1.150	0.578 to 2.287	0.690
PHQ4_1_Prefer not to say	0.003	1.003	0.463 to 2.172	0.994
PHQ4_1_Several days	0.124	1.131	0.570 to 2.247	0.724
PHQ4_2_More than half the days	0.219	1.245	0.693 to 2.239	0.463
PHQ4_2_Nearly every day	0.278	1.321	0.725 to 2.406	0.363
PHQ4_2_Not at all	0.389	1.476	0.827 to 2.634	0.188
PHQ4_2_Prefer not to say	0.654	1.923	0.966 to 3.825	0.063
PHQ4_2_Several days	0.316	1.371	0.769 to 2.444	0.284
PHQ4_3_More than half the days	-0.240	0.786	0.435 to 1.423	0.427
PHQ4_3_Nearly every day	-0.247	0.781	0.428 to 1.425	0.421
PHQ4_3_Not at all	-0.206	0.814	0.452 to 1.465	0.492
PHQ4_3_Prefer not to say	-0.793	0.452	0.226 to 0.904	0.025
PHQ4_3_Several days	-0.210	0.811	0.452 to 1.455	0.482
PHQ4_4_More than half the days	0.096	1.101	0.588 to 2.062	0.764
PHQ4_4_Nearly every day	0.217	1.242	0.658 to 2.344	0.503
PHQ4_4_Not at all	-0.280	0.755	0.407 to 1.402	0.374
PHQ4_4_Prefer not to say	-0.063	0.939	0.452 to 1.951	0.867
PHQ4_4_Several days	-0.017	0.983	0.530 to 1.823	0.956

11.6 Appendix F: Australia-US harmonisation

Source: Generated from `aus_cleaned_data_preprocessing.csv` and `cleaned_data_preprocessing_usa.csv` during the Australia-US harmonisation step, implemented in `5_prepare_combined_aus_us.py`.

11.6.1 Appendix F.1: Common variable set

Table 15: Common core and numeric variables in the harmonised Australia–US dataset.

Construct	Source variable	Coding
Country label	country	Australia or United States
High mask-wearing outcome	mask_binary	0 = comparison category; 1 = high mask-wearing
Sustained mandate-period exposure	within_mandate_period	0 = before/no identified sustained mandate; 1 = during mandate
Age	age	Numeric
Household size	household_size	Numeric; eight or more coded as 8
Health-related survey measure	i2_health	Numeric
Non-mask protective behaviour	protective_behaviour_nomask_scale	Median of available non-mask behaviour items
Fortnight time index	week_number	Consecutive 14-day interval using the country-specific origin
US country indicator	country_US	0 = Australia; 1 = United States
Country-by-mandate interaction	mandate_x_US	Mandate-period indicator multiplied by the US indicator

Table 16: Common categorical indicator variables retained in the harmonised Australia–US dataset.

Construct	Source indicator	Represented response category
Confidence in government	WCRex2_A lot of confidence	A lot of confidence
Confidence in government	WCRex2_Don't know	Don't know
Confidence in government	WCRex2_No confidence at all	No confidence at all
Confidence in government	WCRex2_Not very much confidence	Not very much confidence
Employment status	employment_status_Not working	Not working
Employment status	employment_status_Part time employment	Part time employment
Employment status	employment_status_Retired	Retired
Employment status	employment_status_Unemployed	Unemployed
Gender	gender_Male	Male
Government response evaluation	WCRex1_Somewhat badly	Somewhat badly
Government response evaluation	WCRex1_Somewhat well	Somewhat well
Government response evaluation	WCRex1_Very badly	Very badly
Government response evaluation	WCRex1_Very well	Very well
PHQ-4 item 1	PHQ4_1_Nearly every day	Nearly every day
PHQ-4 item 1	PHQ4_1_Not at all	Not at all
PHQ-4 item 1	PHQ4_1_Prefer not to say	Prefer not to say
PHQ-4 item 1	PHQ4_1_Several days	Several days
PHQ-4 item 2	PHQ4_2_Nearly every day	Nearly every day
PHQ-4 item 2	PHQ4_2_Not at all	Not at all
PHQ-4 item 2	PHQ4_2_Prefer not to say	Prefer not to say
PHQ-4 item 2	PHQ4_2_Several days	Several days
PHQ-4 item 3	PHQ4_3_Nearly every day	Nearly every day
PHQ-4 item 3	PHQ4_3_Not at all	Not at all
PHQ-4 item 3	PHQ4_3_Prefer not to say	Prefer not to say

Construct	Source indicator	Represented response category
PHQ-4 item 3	PHQ4_3_Several days	Several days
PHQ-4 item 4	PHQ4_4_Nearly every day	Nearly every day
PHQ-4 item 4	PHQ4_4_Not at all	Not at all
PHQ-4 item 4	PHQ4_4_Prefer not to say	Prefer not to say
PHQ-4 item 4	PHQ4_4_Several days	Several days
Previous health status	i9_health_Not sure	Not sure
Previous health status	i9_health_Yes	Yes
Vaccination willingness	i11_health_Not sure	Not sure
Vaccination willingness	i11_health_Somewhat unwilling	Somewhat unwilling
Vaccination willingness	i11_health_Somewhat willing	Somewhat willing
Vaccination willingness	i11_health_Very unwilling	Very unwilling
Vaccination willingness	i11_health_Very willing	Very willing

11.6.2 Appendix F.2: Country-specific sample composition

Source: Generated from `aus_cleaned_data_preprocessing.csv` and `cleaned_data_preprocessing_usa.csv` during the Australia-US harmonisation step, implemented in `5_prepare_combined_aus_us.py`.

Table 17: Composition of the harmonised Australia–US dataset.

Country	Total records	High mask-wearing, n	High mask-wearing, %	During sustained mandate, n	During sustained mandate, %
Australia	40,136	21,658	53.96%	25,191	62.76%
US	32,458	20,477	63.09%	22,294	68.69%

11.7 Appendix G: Hold-out robustness results

11.7.1 Appendix G.1: Data partitioning and sample sizes

Source: Generated from `04b_rq2_holdout_test_performance.csv` and `06b_rq3_holdout_test_performance.csv`; these files were produced by `4b_rq2_holdout_robustness_usa.py` and `6b_rq3_holdout_robustness_interaction.py`, respectively.

Table 18: Sample sizes used in the hold-out robustness analyses.

Analysis	Full sample	Training sample	Testing sample
US logistic regression	26,176	20,940	5,236
Australia–US interaction model	72,594	58,075	14,519

11.7.2 Appendix G.2: US model hold-out estimates

Source: Generated from `04c_rq2_holdout_summary_table.csv`, which was produced from the US hold-out robustness analysis implemented in `4b_rq2_holdout_robustness_usa.py`.

Table 19: US mandate-period estimates in the full, training and testing samples.

Sample	Adjusted odds ratio	95% confidence interval	p-value	N
Full sample	1.68	1.54-1.84	<0.001	26,176
Training sample	1.66	1.51-1.83	<0.001	20,940
Testing sample	1.80	1.48-2.19	<0.001	5,236

11.7.3 Appendix G.3: Interaction-model hold-out estimates

Source: Generated from `06c_rq3_holdout_summary_table.csv`, which was produced from the Australia-US interaction hold-out robustness analysis implemented in `6b_rq3_holdout_robustness_interaction.py`.

Table 20: Key estimates from the Australia-US interaction model in the full, training and testing samples.

Sample	Term	Adjusted odds ratio	95% confidence interval	p-value	N
Full sample	Mandate period	6.97	6.57-7.39	<0.001	72,594
Full sample	US country indicator	4.85	4.5-5.22	<0.001	72,594
Full sample	Mandate x US	0.21	0.19-0.23	<0.001	72,594
Training sample	Mandate period	7.01	6.56-7.48	<0.001	58,075
Training sample	US country indicator	4.95	4.56-5.39	<0.001	58,075
Training sample	Mandate x US	0.21	0.19-0.23	<0.001	58,075
Testing sample	Mandate period	6.92	6.06-7.9	<0.001	14,519
Testing sample	US country indicator	4.48	3.79-5.3	<0.001	14,519
Testing sample	Mandate x US	0.23	0.19-0.27	<0.001	14,519